# PRACHINBURI DISTRICT 1 CAMPAIGN DASHBOARD

แดชบอร์ดวิเคราะห์ข้อมูลเลือกตั้ง · ปราจีนบุรี เขต 1

Streamlit | PyDeck | GeoPandas | Google Cloud Storage | Docker

| 129K+ | 76% | 7 | 2 |
|---|---|---|---|
| Eligible Voters | Turnout | Sub-districts | Deployed Apps |

# The Challenge

CAMPAIGN TEAMS NEED REAL-TIME SPATIAL INTELLIGENCE — NOT SPREADSHEETS

📊

### Scattered Data

Election results, polling stations, and field notes lived in separate files with no unified view.

🗺️

### No Spatial Context

Voting patterns varied by geography but could not be visualized at the sub-district level.

📋

### Manual Tracking

Field visit records and campaign pin locations were managed in paper or unstructured notes.

🔒

### No Access Control

Sensitive data needed to be shared publicly for some users and restricted for others.

# The Solution

A UNIFIED, INTERACTIVE WEB DASHBOARD WITH TWO ACCESS TIERS

### Admin Dashboard

Full access: edit sub-district colors, add comments, manage visit records, upload KML overlays.

### Public Read-Only App

Simplified view for public access — map + election results, no sensitive data exposed.

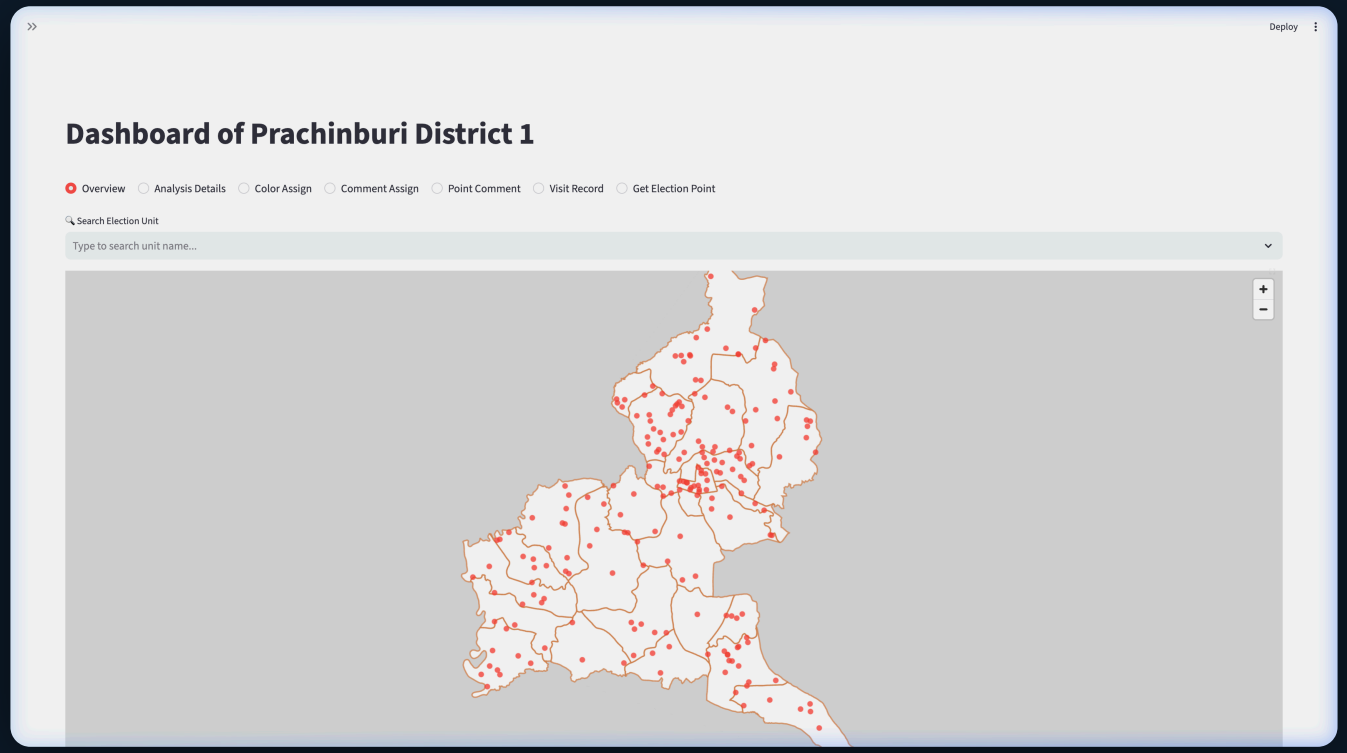
### Cloud Persistence

Google Cloud Storage syncs all state (comments, visits, colors) across users in real time.

### Mobile Optimized

Responsive layout designed for field workers on phones, not just desktop browsers.

# Interactive Map View



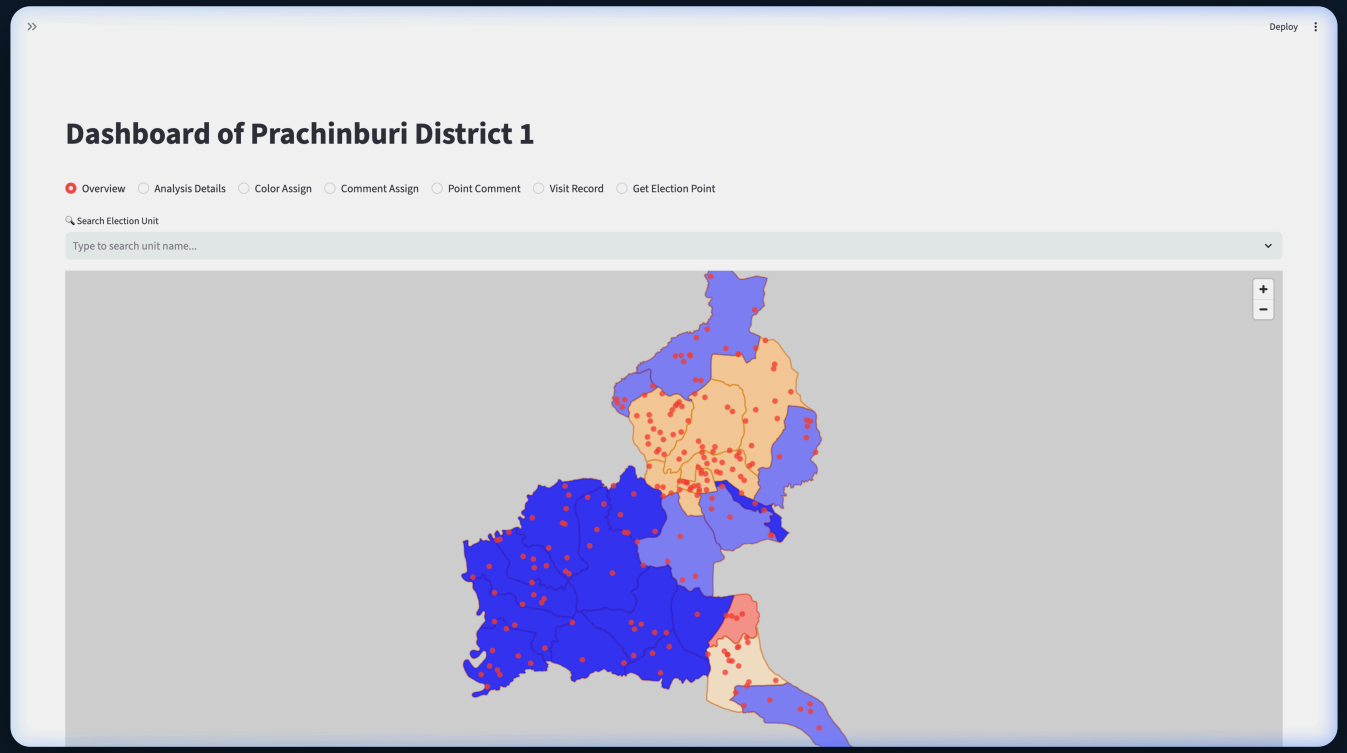

🗺️ Election unit points

📍 Campaign pin locations

🏘️ Sub-district boundaries

🔍 Search by unit name

# Map: Winners Layer



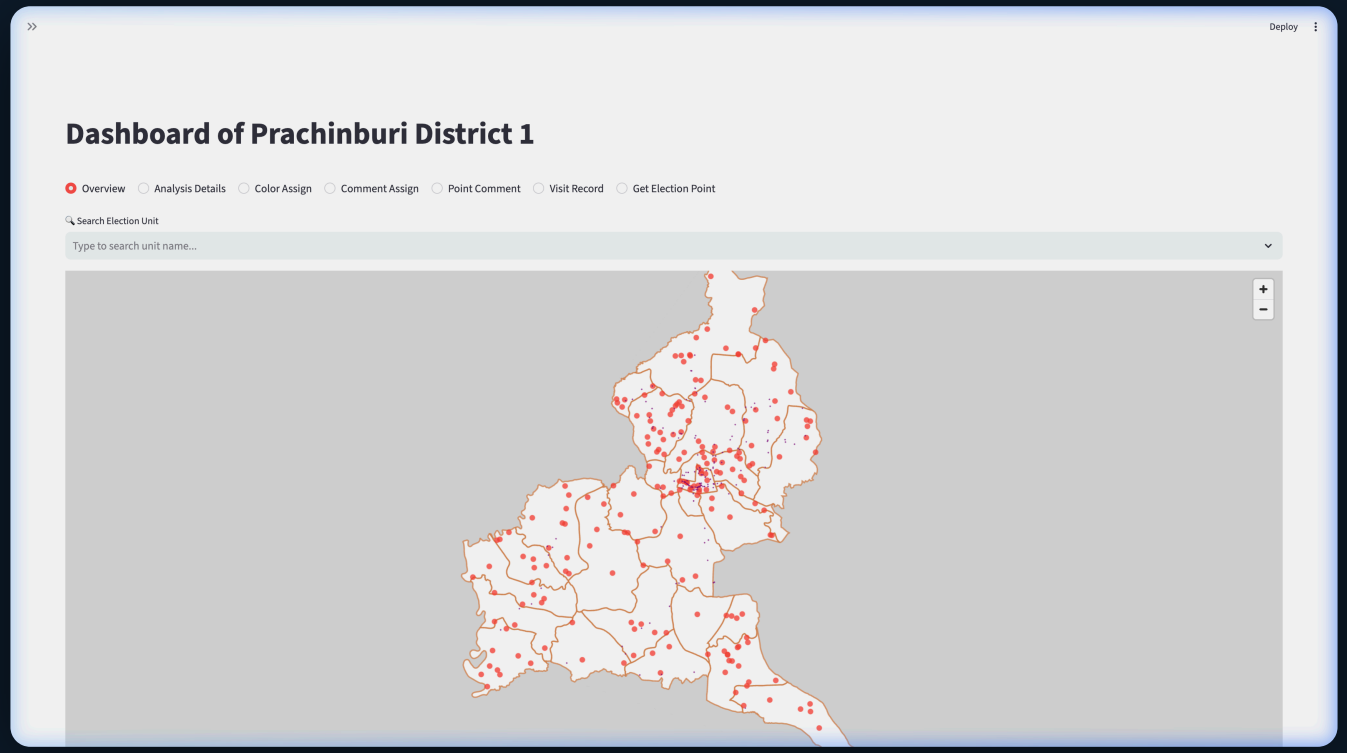

🟠 ก้าวไกล (Orange)

🔵 ภูมิใจไทย (Blue)

🔴 เพื่อไทย (Red)

🏆 Winning party by sub-district

# Map: Election Unit Points

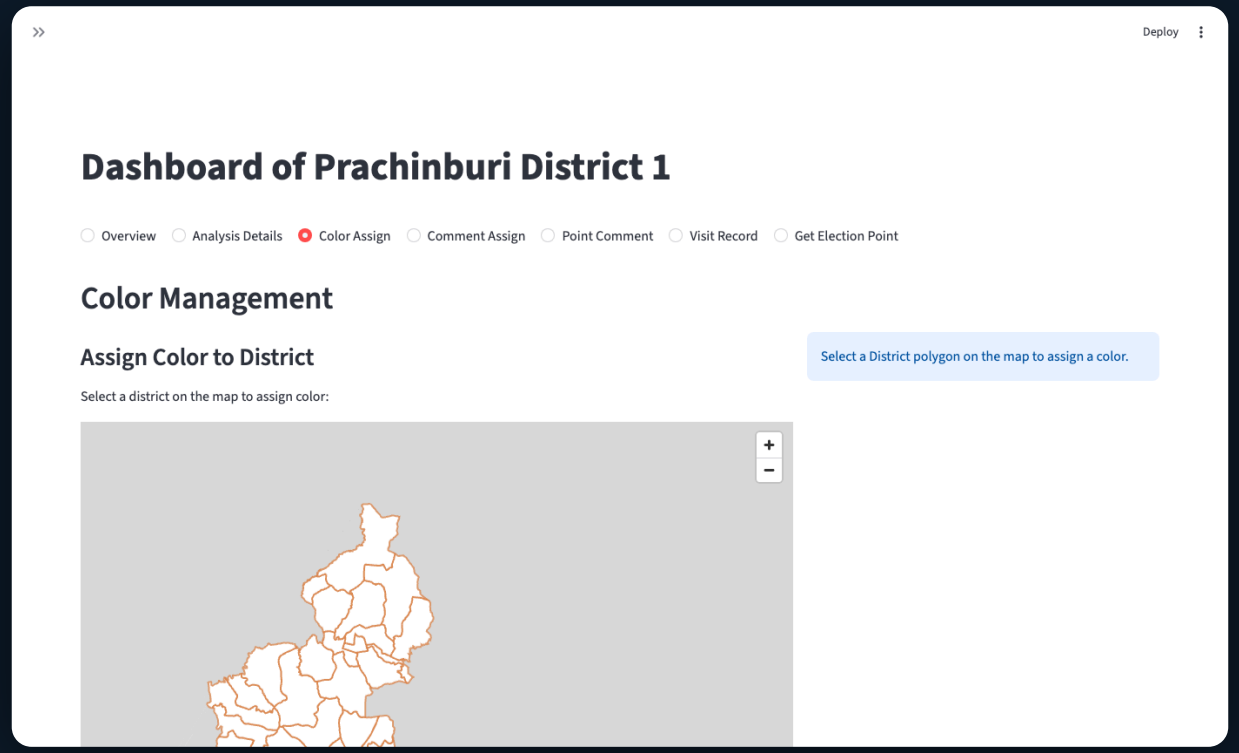



📍 All polling station locations

🏘 Sub-district boundaries

🔍 Searchable by unit name

# Color Assignment




🎨 Assign strategic zones by color

🖱 Click district on map to assign

🔄 Synced to GCS in real time

# Analysis Details

Deploy

## Dashboard of Prachinburi District 1

Overview | Analysis Details | Color Assign | Comment Assign | Point Comment | Visit Record | Get Election Point

### Analysis Details: Aggregated by Sub-district (ตำบล)

| | ตำบล | ผู้มีสิทธิ์ | ผู้มาใช้สิทธิ์ | เปอร์เซ็นต์ใช้สิทธิ์ | บัตรเสีย | ไม่เลือกผู้ใด | ก้าวไกล | ชาติพัฒนากล้า | ชาติไทยพัฒนา | ประชาชาติ | ประชาธิปัตย์ | พลังประชารัฐ | ภูมิใจไทย | รวมไทยสร้างชาติ | เพื่อไทย | เสรีรวมไทย | ไทยสร้างไทย | Winner | Winner_Votes | Winner_Pct |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | กระทุ่มแพ้ว | 1157 | 935 | 80.8124 | 35 | 14 | 188 | 0 | 0 | 0 | 3 | 9 | 529 | 11 | 139 | 3 | 2 | ภูมิใจไทย | 529 | 56.5775 |
| 1 | คู้ลำพัน | 913 | 682 | 74.6988 | 32 | 14 | 115 | 0 | 0 | 0 | 0 | 13 | 236 | 8 | 250 | 3 | 6 | เพื่อไทย | 250 | 36.6569 |
| 2 | ดงขี้เหล็ก | 8681 | 6839 | 78.7812 | 293 | 227 | 2510 | 0 | 0 | 0 | 35 | 232 | 2228 | 234 | 960 | 41 | 52 | ก้าวไกล | 2510 | 36.7013 |
| 3 | ดงพระราม | 12056 | 9034 | 74.9336 | 501 | 411 | 3020 | 0 | 0 | 0 | 48 | 1153 | 2344 | 396 | 958 | 68 | 80 | ก้าวไกล | 3020 | 33.4293 |
| 4 | ท่างาม | 3843 | 3066 | 79.7814 | 111 | 129 | 931 | 0 | 0 | 0 | 35 | 286 | 990 | 182 | 352 | 19 | 19 | ภูมิใจไทย | 990 | 32.2896 |
| 5 | บางกระเบา | 2845 | 2208 | 77.6098 | 74 | 45 | 540 | 0 | 0 | 0 | 3 | 14 | 1034 | 47 | 400 | 20 | 16 | ภูมิใจไทย | 1034 | 46.8297 |
| 6 | บางขาม | 1435 | 1137 | 79.2334 | 31 | 22 | 264 | 0 | 0 | 0 | 1 | 6 | 630 | 17 | 151 | 5 | 6 | ภูมิใจไทย | 630 | 55.409 |
| 7 | บางบริบูรณ์ | 1015 | 816 | 80.3941 | 27 | 18 | 182 | 0 | 0 | 0 | 2 | 72 | 421 | 19 | 65 | 2 | 6 | ภูมิใจไทย | 421 | 51.5931 |
| 8 | บางปลาร้า | 1829 | 1425 | 77.9114 | 53 | 29 | 331 | 0 | 0 | 0 | 6 | 16 | 769 | 16 | 187 | 4 | 10 | ภูมิใจไทย | 769 | 53.9649 |
| 9 | บางพลวง | 4446 | 3349 | 75.3261 | 130 | 64 | 719 | 0 | 0 | 0 | 7 | 59 | 1851 | 102 | 369 | 21 | 15 | ภูมิใจไทย | 1851 | 55.2702 |

Total Eligible Voters

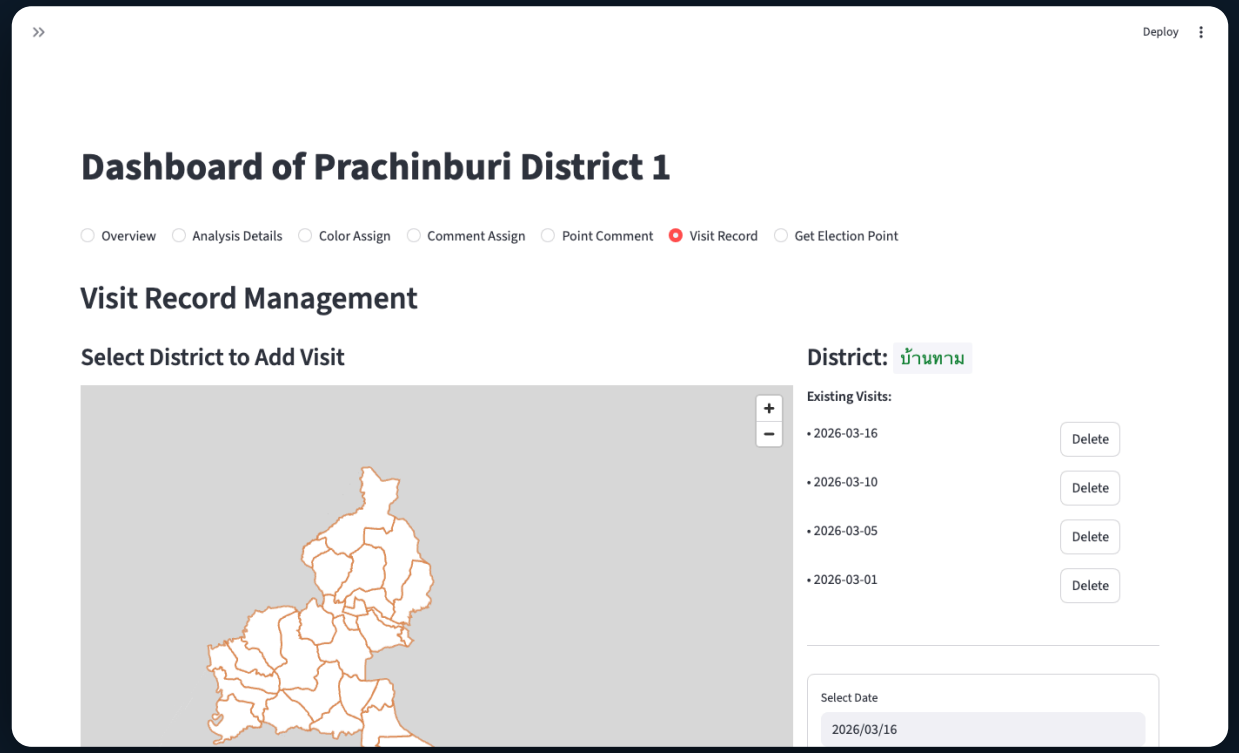
129,146

Total Turnout

98,727

🥇 Winner per sub-district | 📈 Vote percentages | 🗳️ Party breakdown | 📊 129K eligible voters

# Visit Record Management




📅 Thai date formatting

🖱 Click-to-select on map

☁ Synced via GCS

✏ Add / edit / delete

# Technology Stack

🐍
**Python 3.9+**
Core language

⚡
**Streamlit**
Web framework

🗺️
**PyDeck**
WebGL maps

🌍
**GeoPandas**
Spatial data

☁️
**Google Cloud Storage**
Persistence layer

🐳
**Docker**
Containerization

# System Architecture

**Data Sources**
CSV · KML · JSON

→

**Streamlit App**
app.py / public_app.py

↔

**Google Cloud Storage**
Comments · Visits · Colors

**Admin User**
Auth via YAML · full write access

→

**Docker Container**
deploy.sh / deploy_public.sh

→

**Public User**
Read-only · no login required

- KML boundary files parsed with `lxml` and overlaid as GeoJSON polygon layers
- Authentication handled by `streamlit-authenticator` with hashed passwords in YAML
- All mutable state (colors, visits, comments) round-trips through GCS for multi-session consistency

# Key Outcomes

⚡

### Faster Decision Making

Campaign team can identify strong/weak sub-districts at a glance instead of querying spreadsheets.

🎯

### Targeted Campaigning

Color-coded sub-district assignments let team coordinators plan routes and resource allocation.

📡

### Real-time Field Sync

Field staff update visit records from mobile; all team members see changes instantly via GCS.

🔒

### Public Transparency

Separate public app allows voters to explore election results without exposing campaign internals.

# Thank You

Built to empower data-driven decisions on the ground.

Prachinburi District 1 · ปราจีนบุรี เขต 1

github.com/sunsunskibiz/prachin-district-one

Streamlit | PyDeck | GeoPandas | GCS | Docker